

PAULSON ALEX FERNANDES

CYBERSECURITY & DIGITAL FORENSICS | SOFTWARE + AI SECURITY

+91 8459150386 | paulsonfernandes.dev@gmail.com | portfolio.paulsonfernandes.tech | linkedin.com/in/paulson-alex-fernandes-754710378 | github.com/ScriptSynapse

PROFILE

B.Tech Computer Science student specializing in Cybersecurity and Forensics, focused on the intersection of cybersecurity, software engineering, and AI. Hands-on experience building security tooling and conducting penetration-testing, network-analysis, threat-intelligence, and malware-analysis labs. Interested in threat detection, cloud security, security engineering, and AI-assisted security solutions.

SELECTED PROJECTS

AI Malware Detection System | *Python, Flask, Random Forest, XGBoost, SQLite, ReportLab*

- Built static Windows executable analysis without execution using 50+ structural, entropy, import/export, and section-level features; added model comparison, explainability, scan history, and downloadable PDF security reports.

HoneyWatch – SSH Threat Intelligence Honeypot | *Python, Paramiko, asyncio, SQLite, WebSockets, GeoIP*

- Built an SSH honeypot with stateful fake-shell emulation, canary credentials, GeoIP enrichment, attack-pattern detection, and malware capture; added real-time WebSocket monitoring and attacker-origin visualization.

VaultIX – Secure Password Manager | *JavaScript, Web Crypto API, AES-256-GCM, PBKDF2*

- Developed a browser-based password vault with client-side AES-256-GCM encryption, PBKDF2 key derivation, cryptographically random password generation, entropy-based strength scoring, auto-lock, and data import/export.

TECHNICAL SKILLS

Security: Penetration Testing, Network Security, Threat Analysis, SIEM, Cloud Security, Malware Analysis

Programming: Python, C++, JavaScript, SQL

AI / ML: Scikit-learn, XGBoost, Random Forest, Anomaly Detection

Security & Dev Tools: Wireshark, Splunk, Zeek, Metasploit, Nmap, John the Ripper, Cain & Abel, Ettercap, Git, Linux

Security Platforms: Microsoft Defender, Microsoft Sentinel, Windows Firewall

EXPERIENCE

Btech Walleh | *Cybersecurity Intern* | Aug 2025

- Investigated Conti ransomware using Splunk, uncovering persistence, credential theft, exploited CVEs, and remediation recommendations.
- Analyzed a Lumma Stealer campaign with Wireshark and Zeek to identify malicious domains, C2 activity, and exfiltration indicators.
- Performed Windows penetration testing with Metasploit/EternalBlue, privilege escalation, and password-hash cracking.
- Simulated ARP poisoning and configured Windows Firewall rules to demonstrate unencrypted-traffic risks and endpoint defense.

Smart India Hackathon 2025 | *Liaison Officer (Internal)* | Sep 2025

- Coordinated communication between participating teams, mentors, and the organizing committee during the internal hackathon.

EDUCATION

MIT World Peace University, Pune | B.Tech, Computer Science & Engineering (Cybersecurity and Forensics) | Jul 2025 – Jun 2029

CERTIFICATIONS

Palo Alto Networks Academy: Cybersecurity Fundamentals; Network Security Fundamentals | Deloitte Australia: Cyber Job Simulation | Cybersecurity Course Completion Certificate

LANGUAGES

English (Full Professional) | Hindi (Native/Bilingual) | Konkani (Native/Bilingual)